

# Stability analysis and solitary waves on Plateau border flows

Benjamin Walpole<sup>1</sup> and Emilian Ionică Părau<sup>1</sup> 

<sup>1</sup>School of Engineering, Mathematics & Physics, University of East Anglia, Norwich Research Park Norwich, NR4 7TJ, UK

**Corresponding author:** Emilian Ionică Părau, [e.parau@uea.ac.uk](mailto:e.parau@uea.ac.uk)

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We investigate the linear stability of a Plateau border and the existence of solitary waves. Firstly, we formulate a new non-orthogonal coordinate system that describes the specific geometry of a Plateau border. Within the framework of this coordinate system, the equations of motion for the fluid potential and free surface are derived. By performing a linear stability analysis we find that the Plateau border is stable under small perturbations. Next, a weakly nonlinear theory is developed, leading to a Korteweg–de Vries equation for the free surface profile. Our weakly nonlinear evolution equation predicts depression solitary waves, such as those observed by Bouret *et al.* (*Phys. Rev. Fluids.*, 2016, vol. 1 no. 4, p. 043902).

**Key words:** capillary waves, solitary waves

## 1. Introduction

Solitary-wave solutions have been extensively investigated in the context of water-wave problems. Elevation gravity solitary waves were first observed by Scott Russell on the Edinburgh–Glasgow Union canal in 1834 (Russell 1845). Since then, solitary waves have been observed both in nature and in laboratory experiments, and have been analysed using asymptotic and numerical methods (see, e.g. (Dauxois & Peyrard 2006) for a recent review). Depression gravity–capillary solitary waves were observed by (Falcon, Laroche & Fauve 2002) on the surface of a thin layer of mercury. Another class of gravity–capillary solitary waves with damped oscillations has also been reported (see Longuet-Higgins 1989; Vanden-Broeck & Dias 1992). More recently, solitary waves have been observed experimentally and investigated numerically on liquid ferrofluid cylinders (see, for example, Blyth & Părau 2014).

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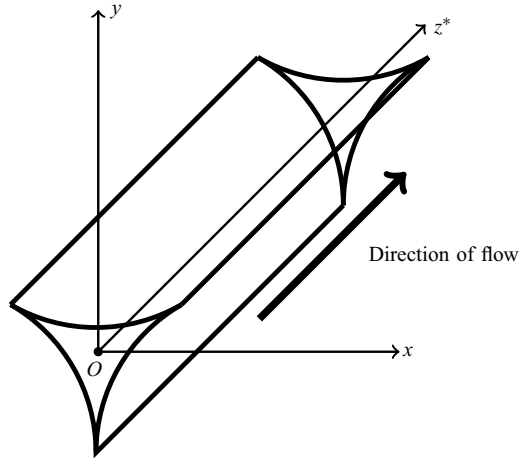


Figure 1. Illustration of Plateau border.

Solitary-wave solutions were also predicted to propagate on Plateau borders (see Argentina *et al.* 2015), which are wall-free liquid microchannels that format at the junction between three bubbles or films within aqueous foams (Cantat *et al.* 2013; Bouret *et al.* 2016). Foams are widely used in domestic and industrial applications, largely due to their high interfacial area (Whittaker & Cox 2019). At equilibrium, soap films meet in threes according to Plateau's laws, creating a microchannel known as the Plateau border (Argentina *et al.* 2015). The geometry of a Plateau border is very specific, as illustrated in figure 1.

In the experiments of Cohen *et al.* (2014, 2015) Argentina *et al.* (2015) and Bouret *et al.* (2016) the Plateau border is inscribed within a triangular prism, and its cross-section was approximated by the tangential contact of three arcs of circles of identical radius of curvature (see also Princen 1986). A range of physical phenomena was observed, including capillary hydraulic jumps, solitary waves, drop coalescence and inertial mass transport in Plateau borders. In particular, Cohen *et al.* (2014) released droplets from above to coalesce with the Plateau border, generating capillary hydraulic jumps that propagated along the channel. Bouret *et al.* (2016) injected bubbles into Plateau borders and subsequently burst them to create depression solitary waves (constrictions) that travelled at constant speed.

Typical physical parameters of the fluids used in these experiments are: viscosity  $\bar{\eta} \sim 10^{-3} \text{ N s m}^{-2}$ , surface tension  $\gamma \sim 4 \times 10^{-2} \text{ N m}^{-1}$  and density  $\rho \sim 10^3 \text{ kg m}^{-3}$ . The typical radius of curvature  $a$  of the Plateau borders was  $\sim 5 \times 10^{-4} \text{ m}$  (ranging from  $3 \times 10^{-4}$  to  $1.8 \times 10^{-3} \text{ m}$ ), while their characteristic length ranged between  $2.5 \times 10^{-2}$  and  $10^{-1} \text{ m}$ . The observed propagation speed  $c$  of solitary waves (see Bouret *et al.* 2016) was  $\sim 0.2 \text{ m s}^{-1}$ . The corresponding non-dimensional parameters are the Reynolds number  $Re$ , the Weber number  $We$  and the Ohnesorge number

$$Re = \frac{\rho c a}{\bar{\eta}} \approx 100, \quad We = \frac{\rho c^2 a}{\gamma} \approx 0.5, \quad Oh = \frac{\sqrt{We}}{Re} \approx 0.007. \quad (1.1)$$

Cohen *et al.* (2014) and Argentina *et al.* (2015) concluded that, when the Ohnesorge number satisfies  $Oh = (\sqrt{We})/(Re) < 0.05$ , the flow is inertia-dominated, allowing waves and hydraulic jumps to propagate along the Plateau border. When  $Oh = (\sqrt{We})/(Re) > 0.05$ , viscous effects dominate and perturbations are strongly damped. In the present study, we focus on the inertia-dominated regime and leave the viscous case for a future investigation.

Previous modelling works investigating depression solitary waves and capillary jumps on Plateau borders (Argentina *et al.* 2015; Bouret *et al.* 2016; Le Doudic *et al.* 2021) have employed shallow-water approximations to derive nonlinear Partial Differential Equations governing the evolution of the free surface. It is also worth noting that, in the literature, some authors have used a cylindrical coordinate system to represent the Plateau-border geometry (see, for example, (Nguyen 2002)). In the present study, we propose to perform a linear stability of a Plateau border subject to small perturbations, followed by the derivation of weakly nonlinear equations for waves propagating along the Plateau border, starting from the full Euler equations written in an alternative coordinate system. In particular a Korteweg–de Vries (KdV) equation will be derived to describe solitary waves, under the slender jet assumption. The long-wave approximation is justified by experimental observations showing that disturbances travelling along the Plateau border vary over length scales much larger than the local radius of curvature. This approach complements previous modelling and experimental studies.

This paper is organised as follows. Section 2 formulates the governing equation in the new non-orthogonal coordinate system. Section 3 examines the linear stability of a Plateau border to perturbations. In § 4 we derive a weakly nonlinear evolution equation for long waves on a Plateau border and a discussion is presented in § 4.

## 2. Problem formulation

### 2.1. Governing equations and boundary conditions

Consider surface-tension-driven flow of an incompressible liquid in a Plateau border with constant density  $\rho$ , neglecting the effects of gravity. As mentioned in the Introduction, measurements obtained by Bouret *et al.* (2016), found that the Reynolds number is large, approximately 100, hence we will assume here that the fluid is inviscid. The time scale of interest is much smaller than time scale of drainage of fluid out of the Plateau border and into the thin films (Koehler *et al.* 2004a,b); therefore, we will neglect the flow out of the Plateau border and into the thin films.

We assume the flow is irrotational so that the fluid velocity is given by  $\mathbf{u} = \nabla\phi$ , where  $\phi(x, y, z^*, t)$  is the fluid potential,  $(x, y, z^*)$  denote Cartesian coordinates and  $t$  is time. The  $z^*$ -axis is oriented along the direction of the flow, while the  $x$ - and  $y$ -axes lie in the transverse cross-section of the Plateau border. The  $y$ -axis is chosen to be normal to one of the Plateau-border surfaces at the mid-point and directed outward.

The representation of the velocity field as a gradient of the fluid potential substituted into the incompressibility condition ( $\nabla \cdot \mathbf{u} = 0$ ), yields Laplace's equation

$$\nabla^2\phi \equiv \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^{*2}} = 0 \quad \text{for } x, y, z^* \in \Omega, \quad (2.1)$$

where  $\Omega$  represents the region of the Plateau border that will be defined further on. Integrating the Euler equations of motion and employing a standard vector calculus identity leads to the Bernoulli equation

$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} = B(t) \quad \text{for } x, y, z^* \in \Omega, \quad (2.2)$$

where  $p = p(x, y, z^*, t)$  is the fluid pressure and  $B(t)$  is the Bernoulli constant, which in general depends on time. Since the flow is surface-tension-driven, we also have the Young–Laplace equation (Batchelor 1967)

$$p - p_0 = \gamma(\nabla \cdot \hat{\mathbf{n}}), \quad \text{for } x, y, z^* \in \partial\Omega, \quad (2.3)$$

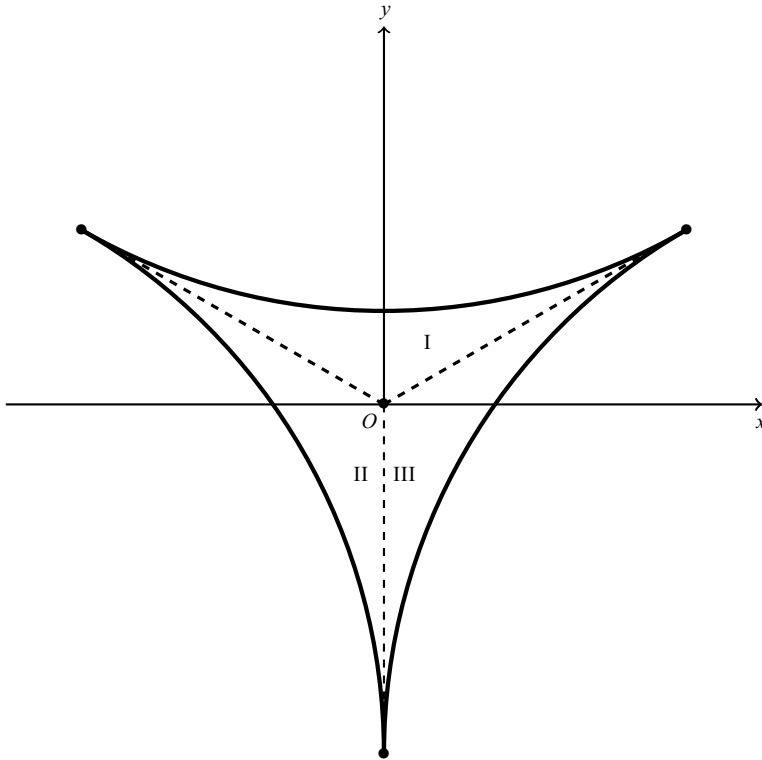


Figure 2. Cross-section of a Plateau border centred at the origin  $O$ .

where  $p_0$  is the assumed constant air pressure outside the Plateau border which can be taken to be zero without loss of generality,  $\partial\Omega$  is the unknown boundary of the Plateau border to be determined,  $\gamma$  is the constant surface tension coefficient and  $\hat{\mathbf{n}}$  is the outward-pointing unit normal to the free surface. Since both the Bernoulli equation and the Young–Laplace equation are evaluated at the boundary of  $\Omega$  (i.e. the free surface), we can eliminate pressure from the Bernoulli equation, to yield

$$\frac{\partial\phi}{\partial t} + \frac{1}{2}|\nabla\phi|^2 + \frac{\gamma}{\rho}(\nabla \cdot \hat{\mathbf{n}}) = B(t) \quad \text{for } x, y, z^* \in \partial\Omega. \quad (2.4)$$

A further boundary condition is required, the kinematic condition, which enforces that the normal velocity of the free surface is equal to the normal component of the fluid velocity at the free surface

$$\nabla\phi \cdot \hat{\mathbf{n}} = \frac{\partial \mathbf{r}}{\partial t} \cdot \hat{\mathbf{n}} \quad \text{for } x, y, z^* \in \partial\Omega, \quad (2.5)$$

where  $\mathbf{r}$  is position vector of a fluid particle on the free surface.

## 2.2. Formulation of the equations of motion in the alternative coordinate system

A cross-section of a single Plateau border can be split into three equally sized regions, namely I, II and III, due to symmetry about the origin  $O$ , as illustrated in [figure 2](#). The region I is bounded by a thick circular arc with radius  $R(z^*, t)$ , centred at  $\nu R(z^*, t)$ . Region I is separated from regions II and III by dashed straight lines on the left- and right-hand sides of the  $y$ -axis, extending from the origin to points where the cusps

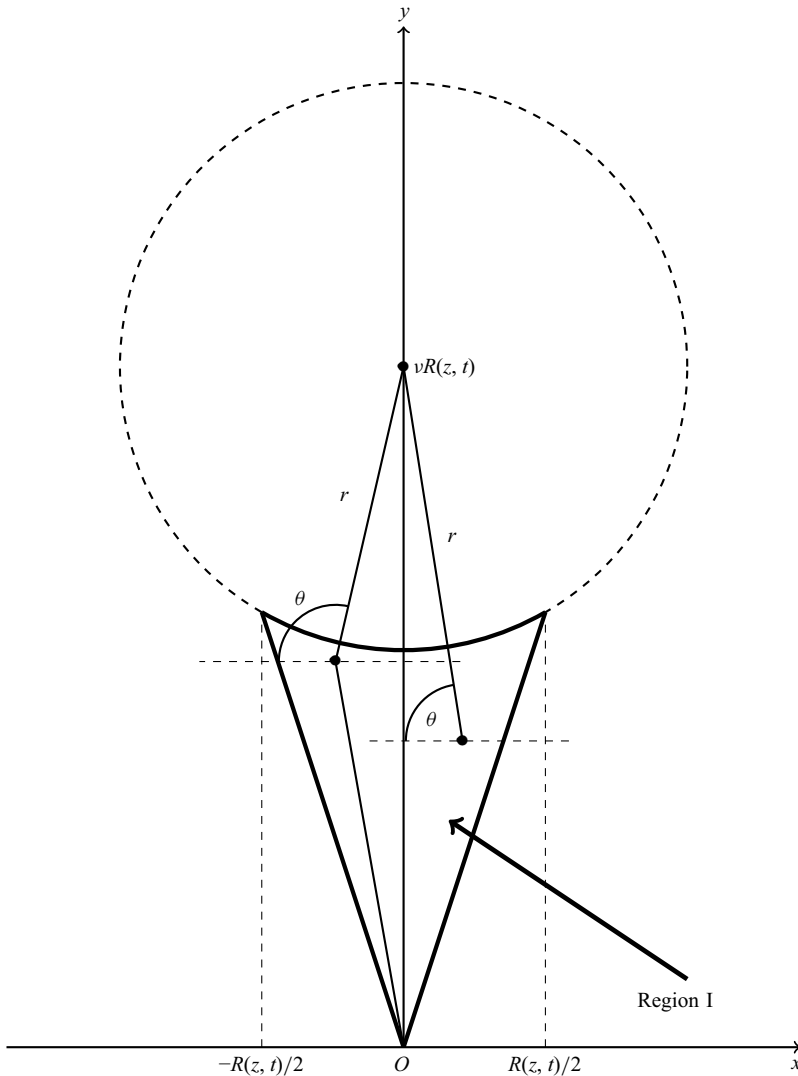


Figure 3. Coordinate system  $(r, \theta)$  in the cross-section of the Plateau border at a fixed  $z$ . Note that the figure is not to scale, being dilated on the  $y$ -axis. Two examples of  $(r, \theta)$  are shown, corresponding to the cases  $\theta < \pi/2$  and  $\theta > \pi/2$ . The positive  $z$ -direction points into the page.

occur, as depicted in [figure 2](#). The flow in regions II and III can be obtained by rotating the solution in region I by  $2\pi/3$  and  $4\pi/3$  counter-clockwise, respectively, about the origin  $O$ .

To describe the geometry of a Plateau border, we extend the work of Argentina *et al.* (2015), which provides a parametrisation of the free surface of a Plateau border. We introduce a new coordinate system  $(r, \theta, z)$  that describes the position of a fluid particle in a Plateau border, where independent variables  $(r, \theta)$  are illustrated in [figure 3](#). The  $z$ -axis points into the page in [figure 3](#). The  $(r, \theta)$  variables describe the location of the fluid particle on a cross-sectional slice of a Plateau border, i.e. at a fixed  $z$ . From [figure 3](#), we observe that

$$x = r \cos \theta, \quad y = vR(z, t) - r \sin \theta, \quad z^* = z, \quad (2.6)$$

where  $\nu = 2/\sqrt{3}$  (see Argentina *et al.* 2015 for details). We note that the origin of this coordinate system depends on  $z$  and  $t$ . The variable  $r$  measures the distance from the point  $\nu R(z, t)$  to a point in the Plateau border, and  $\theta$  measures the angle between the horizontal and the line subtended by the joining of the point  $\nu R(z, t)$  to a point in the Plateau border.

We observe that the free surface is independent of  $\theta$  so that

$$r = R(z, t), \quad (2.7)$$

defines the location of the free surface. We note that when  $r = R(z, t)$ , (2.6) coincides with the parametrisation of the free surface of a Plateau border as given in Argentina *et al.* (2015). From figure 3, it follows that the Cartesian variables take on the following ranges inside the Plateau border region we are considering (i.e. region I):

$$-\frac{R(z^*, t)}{2} \leq x \leq \frac{R(z^*, t)}{2}, \quad \frac{\sqrt{3}}{3}|x| \leq y \leq \nu R(z^*, t) - \sqrt{R^2(z^*, t) - x^2}, \quad -\infty < z^* < \infty. \quad (2.8)$$

The inequalities (2.8) define region I of a Plateau border, and therefore,  $\Omega$ . From (2.6) and (2.8), it follows that  $(r, \theta, z)$  take on the following values inside region I of the Plateau border:

$$R(z, t) \leq r \leq \nu R(z, t), \quad \frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}, \quad -\infty < z < \infty. \quad (2.9)$$

We define the  $(r, \theta, z)$  in the coordinate system for the Plateau border as

$$r = \sqrt{x^2 + (y - \nu R(z^*, t))^2}, \quad (2.10)$$

$$\theta = \begin{cases} \arctan\left(\frac{\nu R(z^*, t) - y}{x}\right) & : x > 0, \\ \frac{\pi}{2} & : x = 0, \\ \pi + \arctan\left(\frac{\nu R(z^*, t) - y}{x}\right) & : x < 0, \end{cases} \quad (2.11)$$

$$z = z^*, \quad (2.12)$$

where  $r > 0$  since it is a measure of distance. We note that  $\nu R(z^*, t) - y > 0$  in region I of the Plateau border, and thus from (2.11) it can be seen that  $\theta$  is continuous at  $x = 0$ . Moreover, it can be shown that  $\theta$  is differentiable at  $x = 0$  by considering limits.

The position,  $\mathbf{r}$ , of a fluid element in the Plateau border is defined by the coordinates  $(r, \theta, z) \equiv (x^1, x^2, x^3)$  as

$$\mathbf{r}(r, \theta, z) = r \cos \theta \hat{\mathbf{i}} + (R(z, t)\nu - r \sin \theta) \hat{\mathbf{j}} + z \hat{\mathbf{k}}, \quad (2.13)$$

where  $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$  are the Cartesian basis vectors pointing in the positive  $x, y, z^*$  directions, respectively. The basis vectors are (see Simmonds 2012)  $\mathbf{e}_i = \partial \mathbf{r} / \partial x^i$ , i.e.

$$\mathbf{e}_1 \equiv \mathbf{e}_r = \frac{\partial \mathbf{r}}{\partial r} = \cos \theta \hat{\mathbf{i}} - \sin \theta \hat{\mathbf{j}}, \quad (2.14)$$

$$\mathbf{e}_2 \equiv \mathbf{e}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = -r \sin \theta \hat{\mathbf{i}} - r \cos \theta \hat{\mathbf{j}}, \quad (2.15)$$

$$\mathbf{e}_3 \equiv \mathbf{e}_z = \frac{\partial \mathbf{r}}{\partial z} = \nu \frac{\partial R}{\partial z} \hat{\mathbf{j}} + \hat{\mathbf{k}}, \quad (2.16)$$

which are neither all orthogonal nor all unit vectors. We note that there is a similarity between the alternative coordinate system  $(r, \theta, z)$  and a cylindrical coordinate system, as seen from (2.14) and (2.15), except  $\mathbf{e}_z \neq \hat{\mathbf{k}}$ . It is also worth noting that they can be normalised as

$$\hat{\mathbf{e}}_r = \mathbf{e}_r, \quad \hat{\mathbf{e}}_\theta = \frac{\mathbf{e}_\theta}{r}, \quad \hat{\mathbf{e}}_z = \frac{\mathbf{e}_z}{\sqrt{\left(v \frac{\partial R}{\partial z}\right)^2 + 1}}. \quad (2.17)$$

The corresponding inverse relations to (2.14)–(2.16) are

$$\hat{\mathbf{i}} = \cos \theta \mathbf{e}_r - \frac{\sin \theta}{r} \mathbf{e}_\theta, \quad (2.18)$$

$$\hat{\mathbf{j}} = -\sin \theta \mathbf{e}_r - \frac{\cos \theta}{r} \mathbf{e}_\theta, \quad (2.19)$$

$$\hat{\mathbf{k}} = v \sin \theta \frac{\partial R}{\partial z} \mathbf{e}_r + \frac{v \cos \theta}{r} \frac{\partial R}{\partial z} \mathbf{e}_\theta + \mathbf{e}_z. \quad (2.20)$$

Using the multi-variable chain rule and (A1) from Appendix A, Laplace's equation (2.1) becomes

$$\begin{aligned} \nabla^2 \phi = & \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} \\ & + \frac{\partial^2 \phi}{\partial r^2} \left( \frac{\partial R}{\partial z} \right)^2 v^2 \sin^2 \theta + \frac{\partial^2 \phi}{\partial r \partial \theta} \left( \frac{\partial R}{\partial z} \right)^2 \frac{2v^2 \cos \theta \sin \theta}{r} + \frac{\partial \phi}{\partial r} \left( \frac{\partial R}{\partial z} \right)^2 \frac{v^2 \cos^2 \theta}{r} \\ & - 2 \frac{\partial \phi}{\partial \theta} \left( \frac{\partial R}{\partial z} \right)^2 \frac{v^2 \sin \theta \cos \theta}{r^2} + \frac{\partial^2 \phi}{\partial \theta^2} \left( \frac{\partial R}{\partial z} \right)^2 \frac{v^2 \cos^2 \theta}{r^2} + 2 \frac{\partial^2 \phi}{\partial z \partial r} \frac{\partial R}{\partial z} v \sin \theta \\ & + \frac{\partial \phi}{\partial r} \frac{\partial^2 R}{\partial z^2} v \sin \theta + 2 \frac{\partial^2 \phi}{\partial z \partial \theta} \frac{\partial R}{\partial z} \frac{v \cos \theta}{r} + \frac{\partial \phi}{\partial \theta} \frac{\partial^2 R}{\partial z^2} \frac{v \cos \theta}{r} = 0, \end{aligned}$$

for  $R(z, t) \leq r \leq vR(z, t)$ ,  $\frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}$ , and  $-\infty < z < \infty$ . (2.21)

The  $|\nabla \phi|^2$  term appearing in the Bernoulli condition (2.4) is more easily calculated in the Cartesian basis vectors. Thus, from (A1), we have

$$\begin{aligned} \nabla \phi = & \frac{\partial \phi}{\partial x} \hat{\mathbf{i}} + \frac{\partial \phi}{\partial y} \hat{\mathbf{j}} + \frac{\partial \phi}{\partial z^*} \hat{\mathbf{k}} \\ = & \left( \cos \theta \frac{\partial \phi}{\partial r} - \frac{\sin \theta}{r} \frac{\partial \phi}{\partial \theta} \right) \hat{\mathbf{i}} + \left( -\sin \theta \frac{\partial \phi}{\partial r} - \frac{\cos \theta}{r} \frac{\partial \phi}{\partial \theta} \right) \hat{\mathbf{j}} \\ & + \left( v \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} + \frac{v \cos \theta}{r} \frac{\partial r}{\partial z} \frac{\partial \phi}{\partial \theta} + \frac{\partial \phi}{\partial z} \right) \hat{\mathbf{k}}. \end{aligned} \quad (2.22)$$

Using (2.22),  $|\nabla \phi|^2$  is given by

$$\begin{aligned} |\nabla \phi|^2 = & \left( \frac{\partial \phi}{\partial r} \right)^2 + \frac{1}{r^2} \left( \frac{\partial \phi}{\partial \theta} \right)^2 + \left( \frac{\partial R}{\partial z} \right)^2 \left( \frac{\partial \phi}{\partial r} \right)^2 v^2 \sin^2 \theta + \left( \frac{\partial R}{\partial z} \right)^2 \left( \frac{\partial \phi}{\partial \theta} \right)^2 \frac{v^2 \cos^2 \theta}{r^2} \\ & + \left( \frac{\partial \phi}{\partial z} \right)^2 + \frac{2v^2 \cos \theta \sin \theta}{r} \left( \frac{\partial R}{\partial z} \right)^2 \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial \theta} + 2v \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial z} \\ & + \frac{2v \cos \theta}{r} \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial \theta} \frac{\partial \phi}{\partial z}. \end{aligned} \quad (2.23)$$

### 2.3. Curvature of the free surface

The material surface that defines the free surface of a Plateau border is

$$F(x, y, z^*, t) \equiv R(z^*, t)^2 - x^2 - (y - vR(z^*, t))^2 = 0. \quad (2.24)$$

Hence the outward-pointing unit normal vector is

$$\hat{\mathbf{n}} = \frac{\nabla F}{|\nabla F|} = \frac{-\cos \theta \hat{\mathbf{i}} + \sin \theta \hat{\mathbf{j}} - \frac{\partial R}{\partial z}(v \sin \theta - 1) \hat{\mathbf{k}}}{\sqrt{1 + \left(\frac{\partial R}{\partial z}\right)^2 (v \sin \theta - 1)^2}}, \quad (2.25)$$

and its divergence is, after some algebra,

$$\nabla \cdot \hat{\mathbf{n}} = \frac{(1 - v \sin \theta) \frac{\partial^2 R}{\partial z^2} R - 1 + (2v \sin \theta - v^2 - 1) \left(\frac{\partial R}{\partial z}\right)^2}{R \left[1 + (v \sin \theta - 1)^2 \left(\frac{\partial R}{\partial z}\right)^2\right]^{3/2}}. \quad (2.26)$$

The Bernoulli condition, after substituting (2.26) and (2.23) into (2.4), is

$$\begin{aligned} \frac{\partial \phi}{\partial t} + \frac{1}{2} \left( \left(\frac{\partial \phi}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial \phi}{\partial \theta}\right)^2 + \left(\frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r}\right)^2 v^2 \sin^2 \theta + \left(\frac{\partial R}{\partial z}\right)^2 \left(\frac{\partial \phi}{\partial \theta}\right)^2 \frac{v^2 \cos^2 \theta}{r^2} \right. \\ \left. + \left(\frac{\partial \phi}{\partial z}\right)^2 + \frac{2v^2 \cos \theta \sin \theta}{r} \left(\frac{\partial R}{\partial z}\right)^2 \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial \theta} + 2v \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial z} + \frac{2v \cos \theta}{r} \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial \theta} \frac{\partial \phi}{\partial z} \right) \\ \left. + \frac{\gamma}{\rho} \left( \frac{(1 - v \sin \theta) \frac{\partial^2 R}{\partial z^2} R - 1 + (2v \sin \theta - v^2 - 1) \left(\frac{\partial R}{\partial z}\right)^2}{R \left[1 + (v \sin \theta - 1)^2 \left(\frac{\partial R}{\partial z}\right)^2\right]^{3/2}} \right) = B(t), \end{aligned} \quad (2.27)$$

on  $r = R(z, t)$ .

### 2.4. Kinematic boundary condition

We note from (2.13) that  $\partial \mathbf{r} / \partial t$ , where  $\mathbf{r}$  is the position of the fluid particle on the free surface, is given by

$$\frac{\partial \mathbf{r}}{\partial t} = \frac{\partial R}{\partial t} \cos \theta \hat{\mathbf{i}} + \frac{\partial R}{\partial t} (v - \sin \theta) \hat{\mathbf{j}}. \quad (2.28)$$

From (2.22), (2.25) and (2.28) the kinematic boundary condition (2.5) becomes, after some algebra,

$$\begin{aligned} \frac{\partial R}{\partial t} (1 - v \sin \theta) - \frac{\partial \phi}{\partial r} \\ + \frac{\partial R}{\partial z} (1 - v \sin \theta) \left( v \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} + \frac{v \cos \theta}{r} \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial \theta} + \frac{\partial \phi}{\partial z} \right) = 0 \quad \text{on } r = R(z, t). \end{aligned} \quad (2.29)$$

### 2.5. Symmetry boundary condition

To establish another boundary condition, consider a cross-sectional slice of a Plateau border. Imagine the fluid velocity vector at the origin  $O$ , pointing radially outwards (in



the  $\mathbf{e}_r$  direction). If the velocity vector has non-zero components in the  $\hat{\mathbf{i}}$  and  $\hat{\mathbf{j}}$  directions, rotating the cross-section of the Plateau border by integer multiples of  $2\pi/3$  would lead to an inconsistency. By symmetry, the fluid velocity of the Plateau border must be identical to the fluid velocity of the rotated Plateau border. The only way to satisfy this for any integer multiple of  $2\pi/3$  is if the components of fluid velocity vector in  $\hat{\mathbf{i}}$  and  $\hat{\mathbf{j}}$  directions at the origin are zero. Mathematically, this means

$$\mathbf{u} \cdot \hat{\mathbf{i}} = 0, \quad \text{and} \quad \mathbf{u} \cdot \hat{\mathbf{j}} = 0, \quad \text{on } r = \nu R(z, t), \quad (2.30)$$

where the origin  $O$  is defined by  $r = \nu R(z, t)$  in region I of the Plateau border. The fluid velocity vector in the alternative coordinate system, expressed in terms of the  $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z\}$  vectors, is

$$\mathbf{u} = u^r \mathbf{e}_r + u^\theta \mathbf{e}_\theta + u^z \mathbf{e}_z, \quad (2.31)$$

where  $u^r, u^\theta, u^z$  are the velocity components in the  $\mathbf{e}_r, \mathbf{e}_\theta$  and  $\mathbf{e}_z$  directions, respectively.

After some work (see [Appendix A](#)) we find that the symmetry condition at  $r = \nu R(z, t)$  is given by

$$u^r \cos \theta - u^\theta r \sin \theta = 0, \quad \text{and} \quad -u^r \sin \theta - u^\theta r \cos \theta + \nu \frac{\partial R}{\partial z} u^z = 0. \quad (2.32)$$

To determine the components of the fluid velocity in terms of the fluid potential, we substitute (2.18)–(2.20) into (2.22), to obtain

$$u^r = \left( 1 + \left( \frac{\partial R}{\partial z} \right)^2 \nu^2 \sin^2 \theta \right) \frac{\partial \phi}{\partial r} + \frac{\nu^2 \cos \theta \sin \theta}{r} \left( \frac{\partial R}{\partial z} \right)^2 \frac{\partial \phi}{\partial \theta} + \nu \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial z}, \quad (2.33)$$

$$u^\theta = \frac{\nu^2 \cos \theta \sin \theta}{r} \left( \frac{\partial R}{\partial z} \right)^2 \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \left( 1 + \left( \frac{\partial R}{\partial z} \right)^2 \nu^2 \cos^2 \theta \right) \frac{\partial \phi}{\partial \theta} + \frac{\nu \cos \theta}{r} \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial z}, \quad (2.34)$$

$$u^z = \nu \sin \theta \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} + \frac{\nu \cos \theta}{r} \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial \theta} + \frac{\partial \phi}{\partial z}. \quad (2.35)$$

The symmetry condition expressed in terms of the fluid potential, after substituting (2.33)–(2.35) into the symmetry condition (2.32), is

$$\frac{\partial \phi}{\partial r} \cos \theta - \frac{1}{r} \frac{\partial \phi}{\partial \theta} \sin \theta = 0, \quad \text{and} \quad \frac{\partial \phi}{\partial r} \sin \theta + \frac{1}{r} \frac{\partial \phi}{\partial \theta} \cos \theta = 0, \quad \text{on } r = \nu R(z, t). \quad (2.36)$$

This completes the derivation of the boundary conditions.

### 3. Linear stability analysis

Following Argentina *et al.* (2015), we perform azimuthal averaging over a given cross-section of the Plateau border. This means we take  $\theta = \pi/2$ . Due to this averaging, we assume the fluid velocity component in the azimuthal direction,  $u^\theta$ , is zero everywhere on  $\theta = \pi/2$ . Furthermore, we assume that all variables are independent of  $\theta$ . This effectively reduces a three-dimensional problem to a two-dimensional one. Henceforth, in this section, we assume

$$\theta = \frac{\pi}{2} \quad \text{and} \quad u^\theta = 0, \quad \frac{\partial}{\partial \theta} = 0 \quad \text{on } \theta = \frac{\pi}{2}. \quad (3.1)$$

We consider the Plateau border to be in a uniform state, meaning the  $r$ -variable is a constant,  $r = a$ . Substituting  $r = a$  into the Young–Laplace (2.3) we find the basic state pressure to be  $p = -\gamma/a$ . We also assume the Plateau border to be stationary, so we define the basic state (denoted with overbars) as

$$\bar{\phi} = 0, \quad \bar{p} = -\frac{\gamma}{a}, \quad \bar{R} = a. \quad (3.2)$$

We perturb this uniform Plateau border by adding small disturbances, scaled by  $\epsilon$ , where  $\epsilon$  represents the characteristic ratio between the amplitude of a wave to its wavelength and we assume  $0 < \epsilon \ll 1$ . Hence, we write

$$\phi(r, z, t) = \epsilon \hat{\phi}(r, z, t) + O(\epsilon^2), \quad R(z, t) = \bar{R} + \epsilon \hat{r}(z, t) + O(\epsilon^2), \quad (3.3)$$

where the hat decoration denote perturbation quantities. Note that, once the solution for  $\hat{\phi}(r, z, t)$  is found, the pressure field can be determined from the Bernoulli equation (2.2). Substituting the exact basic state solution (3.2) into the Bernoulli equation we find the Bernoulli constant to be  $B(t) = -\gamma/(\rho a)$ . We substitute (3.3) into the governing Laplace equation (2.21) and the boundary conditions: the Bernoulli condition (2.27), the kinematic condition (2.29) and the symmetry condition (2.36). After performing Taylor expansions of the boundary conditions about  $r = a$  for Bernoulli and kinematic, and about  $r = av$  for the symmetry condition, and then neglecting terms of  $O(\epsilon^2)$  and higher, we obtain the linearised equations of motion at  $O(\epsilon)$

$$\frac{\partial^2 \hat{\phi}}{\partial r^2} + \frac{1}{r} \frac{\partial \hat{\phi}}{\partial r} + \frac{\partial^2 \hat{\phi}}{\partial z^2} = 0 \quad \text{for } a \leq r \leq av, \quad (3.4)$$

$$\frac{\partial \hat{\eta}}{\partial t} (1 - \nu) - \frac{\partial \hat{\phi}}{\partial r} = 0 \quad \text{on } r = a, \quad (3.5)$$

$$\frac{\partial \hat{\phi}}{\partial t} + \frac{\gamma}{\rho} \left( (1 - \nu) \frac{\partial^2 \hat{\eta}}{\partial z^2} + \frac{\hat{\eta}}{a^2} \right) = 0 \quad \text{on } r = a, \quad (3.6)$$

$$\frac{\partial \hat{\phi}}{\partial r} = 0 \quad \text{on } r = av. \quad (3.7)$$

We seek a solution for these linearised equations of motion that is periodic in  $z$  and exponential in  $t$ . Specifically, we look for solutions of the form

$$\hat{\phi}(r, z, t) = \Phi(r) \exp(ikz + st) + \text{c.c.}, \quad \hat{\eta}(z, t) = \hat{\eta}_0 \exp(ikz + st) + \text{c.c.}, \quad (3.8)$$

where  $\hat{\eta}_0$  is a constant, c.c. denotes the complex conjugate of the preceding part of the equation (this convention applies throughout the rest of the text, but will not be explicitly stated again). Here,  $k$  is the real and positive wavenumber, and  $s$  is the temporal growth rate. If  $s \in i\mathbb{R}$  (purely imaginary), then waves will emerge on the free surface of the Plateau border. If  $s \in \mathbb{R}^{>0}$  (real and positive), the flow is unstable and will diverge from the basic state. If  $s \in \mathbb{R}^{\leq 0}$  (real and non-positive), the flow is stable and, after a sufficiently long time, the perturbations will decay and the flow will return to the basic state.

Substituting (3.8) into (3.4), we find that  $\Phi(r)$  satisfies

$$\frac{d^2 \Phi}{dr^2} + \frac{1}{r} \frac{d\Phi}{dr} - k^2 \Phi = 0, \quad (3.9)$$

which is Bessel's modified equation of zeroth order. It has two linearly independent solutions:  $I_0$  and  $K_0$ , known as the modified Bessel functions of the first and second kinds, both of zeroth order, respectively. It is well known (see, for example, Olver 2010) that modified Bessel function  $I_n$  (modified Bessel function of the first kind of order  $n$ )

is finite at the origin and grow exponentially with a singularity at infinity, while  $K_n$  (modified Bessel function of the second kind of order  $n$ ) is singular at the origin and decays exponentially. Therefore, both  $I_n(kr)$  and  $K_n(kr)$  remain finite within the Plateau border ( $a \leq r \leq av$ ). Thus, (3.9) has the general solution

$$\Phi(r) = C_1 I_0(kr) + C_2 K_0(kr), \quad (3.10)$$

where  $C_1, C_2$  are arbitrary constants of integration. We have three unknown constants  $C_1, C_2$  and  $\hat{\eta}_0$ , and therefore require three boundary conditions. Applying the symmetry condition (2.36) to (3.10), we find

$$\Phi(r) = C_1 \left( I_0(kr) + \frac{I_1(kva)}{K_1(kva)} K_0(kr) \right). \quad (3.11)$$

We express the kinematic (3.5) and Bernoulli conditions (3.6) in matrix-vector form as

$$\begin{pmatrix} -k \left( I_1(ka) - \frac{I_1(kva)}{K_1(kva)} K_1(ka) \right) & s(1-v) \\ s \left( I_0(kr) + \frac{I_1(kva)}{K_1(kva)} K_0(kr) \right) & \frac{\gamma}{\rho} \left( \frac{1}{a^2} - (1-v)k^2 \right) \end{pmatrix} \begin{pmatrix} C_1 \\ \hat{\eta}_0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (3.12)$$

The first and second rows correspond to the kinematic and Bernoulli conditions, respectively. For this matrix-vector (3.12) to admit a non-trivial solution, we require that the determinant of the matrix vanishes, which yields

$$s^2 = -\frac{\gamma(ka)}{\rho a^3} \left[ \frac{1}{1-v} - (ak)^2 \right] \frac{I_1(ka)K_1(kva) - I_1(kva)K_1(ka)}{I_0(ka)K_1(kva) + I_1(kva)K_0(ka)}. \quad (3.13)$$

This is the linear inviscid dispersion relation for a Plateau border with azimuthal averaging. We observe the term in the square brackets in the dispersion relation, (3.13), is negative for all real  $k$ . To analysis the stability of the Plateau border, we need to determine the sign of

$$\Lambda := \frac{I_1(ka)K_1(kva) - I_1(kva)K_1(ka)}{I_0(ka)K_1(kva) + I_1(kva)K_0(ka)}. \quad (3.14)$$

We note the following inequalities for the modified Bessel functions (see, for example, Radwan 1992; Olver 2010, p. 5):

$$|K_1(\xi q)| > |K_1(\xi)|, \quad (3.15)$$

$$I_1(\xi q) < I_1(\xi). \quad (3.16)$$

It is well known that  $I_n(z), K_n(z)$  are positive for non-negative arguments; thus, the denominator of (3.14) is positive for all  $ka \geq 0$ . If we define  $\xi = kva$  and  $q = 1/v$  such that  $0 < q < 1$ , then since

$$\frac{I_1(\xi)K_1(\xi q)}{I_1(\xi q)K_1(\xi)} > 1, \quad (3.17)$$

it follows that the numerator of (3.14) is negative for all  $ka \geq 0$ . Consequently,  $\Lambda < 0$  for all  $ka \geq 0$  which implies  $s \in i\mathbb{R}$  and the Plateau border is stable for all real and positive wavenumbers.

### 3.1. Non-dimensionalisation and long-wave speed

We now derive the non-dimensional dispersion relation by non-dimensionalising all lengths in units of the unperturbed  $r$ -coordinate,  $a$ . We perform the non-dimensionalisation as follows:

$$r = ar', \quad z = az', \quad t = \sqrt{\frac{a^3 \rho}{\gamma}} t', \quad (3.18)$$

where primed quantities denote non-dimensional ones. Substituting (3.18) into (3.13), we obtain the non-dimensional dispersion relation

$$s^2 = -k \left( \frac{1}{1-\nu} - k^2 \right) \frac{I_1(k)K_1(k\nu) - I_1(k\nu)K_1(k)}{I_0(k)K_1(k\nu) + I_1(k\nu)K_0(k)}. \quad (3.19)$$

By writing  $s = -ikc$ , where  $c$  is the phase speed of the wave, and substituting in (3.19) we obtain

$$c^2 = \frac{1}{k} \left( \frac{1}{1-\nu} - k^2 \right) \frac{I_1(k)K_1(k\nu) - I_1(k\nu)K_1(k)}{I_0(k)K_1(k\nu) + I_1(k\nu)K_0(k)}. \quad (3.20)$$

The leading-order terms of the asymptotic expansions for small arguments of the modified Bessel functions are (Olver 2010)

$$I_0(\chi) \sim 1, \quad I_1(\chi) \sim \frac{1}{2}\chi, \quad K_0(\chi) \sim -\log \chi, \quad K_1(\chi) \sim \frac{1}{\chi}, \quad \text{as } \chi \rightarrow 0. \quad (3.21)$$

Substituting these asymptotic expansions (3.21) into the dispersion relation (3.20), we find the long-wave speed and its dimensional counterpart (denoted with  $\star$ ) to be

$$c_0 = \sqrt{\frac{\nu+1}{2}} \approx 1.0308, \quad c_0^\star = \sqrt{\frac{\gamma}{\rho a} \frac{\nu+1}{2}}. \quad (3.22)$$

A plot of the dispersion relation (3.20), showing  $c$  against the wavenumber  $k$ , is depicted in figure 4. From figure 4, we observe the minimum is located at  $k = 0$ , which is confirmed by numerical investigation.

Different types of solitary waves can bifurcate from either the long-wave limit or the minimum phase speed (see Dauxois & Peyrard 2006). However, we have established a minimum phase speed for finite  $k$  does not exist in our model. This suggests that, when we proceed to weakly nonlinear theory, we should use a perturbation expansion for  $k \ll 1$ , i.e. in the long-wave limit.

The absence of a minimum phase speed for finite  $k$  implies that pure solitary waves, such as those observed by Bouret *et al.* (2016), must travel with phase speed slightly less than the long-wave speed. The reasoning behind this is that if the phase speed were slightly larger than the long-wave limit, linear capillary waves would resonate with the fully nonlinear wave, resulting in a generalised solitary wave (Hunter & Vanden-Broeck 1983; Vanden-Broeck 2010) – a solitary wave with non-decaying oscillatory tails. This is indeed confirmed by our weakly nonlinear theory (as discussed in the next section).

Finally, it is important to note that our theoretical predictions differ from those reported in previous studies (Argentina *et al.* 2015; Bouret *et al.* 2016; Le Doudic *et al.* 2021). Our analysis predicts the long-wave speed to be  $\sqrt{\gamma(\nu+1)/(2\rho a)}$ , whereas previous authors reported the long-wave speed as  $\sqrt{\gamma/(2\rho a)}$ . This difference may arise from the fact that in those studies they model a slightly different problem of a Plateau border formed by holding films attached to a triangular prism frame, while in our case we assumed that the

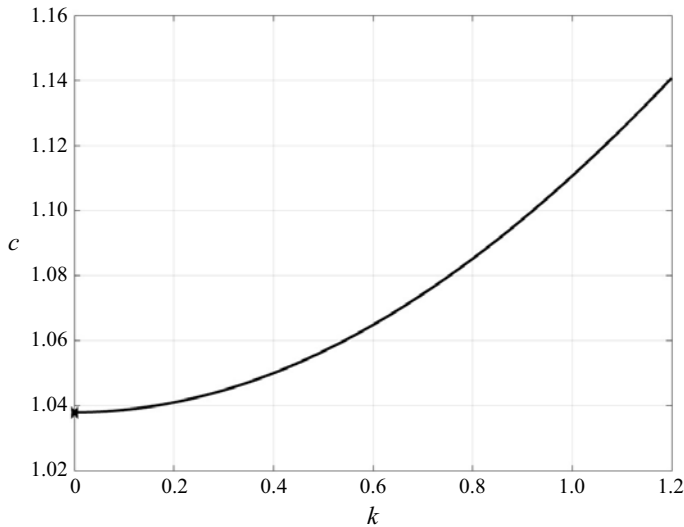


Figure 4. Plot of dispersion relation (3.19): dimensionless phase speed  $c$  against the wavenumber  $k$ .

Plateau border is formed at the intersection of three bubbles or liquid foams. Hence, we neglect the physical effect of any holding film attached to a prism frame.

#### 4. Weakly nonlinear theory and derivation of Korteweg–de Vries equation

##### 4.1. Dimensionless equations of motion

We assume that the characteristic horizontal length scale in the  $z$ -direction, denoted by  $\ell$ , is long compared with the amplitude of the undisturbed free surface of the Plateau border  $r = a$  (where  $a$  is a constant). This allows us to define a slenderness ratio

$$\epsilon = \frac{a}{\ell} \ll 1. \quad (4.1)$$

Following Wang, Papageorgiou & Vanden-Broeck (2015), we non-dimensionalise the problem using the following scalings:

$$\phi = \sqrt{\frac{\gamma \ell^2}{\rho a}} \phi', \quad r = ar', \quad z = \ell z', \quad R = aR', \quad t = \sqrt{\frac{\rho a \ell^2}{\gamma}} t', \quad (4.2)$$

where primes denote non-dimensional quantities. As in Argentina *et al.* (2015) and §3, we average the azimuthal coordinate ( $\theta$ ) over a given cross-section. Henceforth, in this section, we assume the conditions given by (3.1).

Substituting (4.2) into Laplace's equation (2.21), the kinematic condition (2.29), the Bernoulli condition (2.27) and the symmetry condition (2.36), we obtain the following equations (after dropping the primes for brevity): Laplace's equation

$$\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \epsilon^2 \frac{\partial^2 \phi}{\partial z^2} + 2\epsilon^2 v \frac{\partial R}{\partial z} \frac{\partial^2 \phi}{\partial r \partial z} + \epsilon^2 v^2 \frac{\partial^2 \phi}{\partial r^2} \left( \frac{\partial R}{\partial z} \right)^2 v^2 + \epsilon^2 v \frac{\partial^2 R}{\partial z^2} \frac{\partial \phi}{\partial r} = 0, \quad (4.3)$$

for  $R(z, t) \leq r \leq \nu R(z, t)$ , the Bernoulli equation

$$\begin{aligned} \epsilon^2 \frac{\partial \phi}{\partial t} + \frac{1}{2} \left( \left( \frac{\partial \phi}{\partial r} \right)^2 + \epsilon^2 \nu^2 \left( \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} \right)^2 + \epsilon^2 \left( \frac{\partial \phi}{\partial z} \right)^2 + 2\epsilon^2 \nu \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial z} \right) \\ + \frac{(1-\nu)\epsilon^4 \frac{\partial^2 R}{\partial z^2} R - \epsilon^2 - (\nu-1)^2 \epsilon^4 \left( \frac{\partial R}{\partial z} \right)^2}{R \left[ 1 + (\nu-1)^2 \epsilon^2 \left( \frac{\partial R}{\partial z} \right)^2 \right]^{3/2}} + \epsilon^2 = 0 \quad \text{on } r = R(z, t). \end{aligned} \quad (4.4)$$

The kinematic boundary condition is

$$\epsilon^2 \frac{\partial R}{\partial t} (1-\nu) - \frac{\partial \phi}{\partial r} + \epsilon^2 \frac{\partial R}{\partial z} (1-\nu) \left( \nu \frac{\partial R}{\partial z} \frac{\partial \phi}{\partial r} + \frac{\partial \phi}{\partial z} \right) = 0 \quad \text{on } r = R(z, t), \quad (4.5)$$

and the symmetry condition is

$$\frac{\partial \phi}{\partial r} = 0 \quad \text{on } r = \nu R(z, t). \quad (4.6)$$

We now seek a solution to (4.3)–(4.6) for  $0 < \epsilon \ll 1$ . Therefore, we look for a a perturbation solution for small  $\epsilon$  with weakly nonlinear free surface deformation, so we write

$$R(z, t) = 1 + \epsilon^2 \eta(z, t), \quad (4.7)$$

and assume the expansion for the velocity potential is of the form

$$\phi(r, z, t) = \epsilon^2 \phi_0(r, z, t) + \epsilon^4 \phi_1(r, z, t) + \epsilon^6 \phi_2(r, z, t) + O(\epsilon^8). \quad (4.8)$$

We apply Taylor expansions to the symmetry condition (4.6) about  $r = \nu$  and then substitute the expansion for the velocity potential into the resulting equation. This yields the following equations at each order:

$$O(\epsilon^2): \quad \left. \frac{\partial \phi_0}{\partial r} \right|_{r=\nu} = 0, \quad (4.9)$$

$$O(\epsilon^4): \quad \left. \frac{\partial \phi_1}{\partial r} \right|_{r=\nu} + \nu \eta \left. \frac{\partial^2 \phi_0}{\partial r^2} \right|_{r=\nu} = 0, \quad (4.10)$$

$$O(\epsilon^6): \quad \left. \frac{\partial \phi_2}{\partial r} \right|_{r=\nu} + \nu \eta \left. \frac{\partial^2 \phi_1}{\partial r^2} \right|_{r=\nu} + \frac{\nu^2 \eta^2}{2} \left. \frac{\partial^3 \phi_0}{\partial r^3} \right|_{r=\nu} = 0. \quad (4.11)$$

#### 4.2. Asymptotic solutions of Laplace's equation

Substituting the perturbation expansion for the velocity potential (4.8) into Laplace's equation (4.3), and collecting terms in a series of  $\epsilon^2$ , we find at each order

$$O(\epsilon^2): \quad \frac{\partial^2 \phi_0}{\partial r^2} + \frac{1}{r} \frac{\partial \phi_0}{\partial r} = 0, \quad (4.12)$$

$$O(\epsilon^4): \quad \frac{\partial^2 \phi_1}{\partial r^2} + \frac{1}{r} \frac{\partial \phi_1}{\partial r} = -\frac{\partial^2 \phi_0}{\partial z^2}, \quad (4.13)$$

$$O(\epsilon^6): \quad \frac{\partial^2 \phi_2}{\partial r^2} + \frac{1}{r} \frac{\partial \phi_2}{\partial r} = -\frac{\partial^2 \phi_1}{\partial z^2} - 2\nu \frac{\partial \eta}{\partial z} \frac{\partial^2 \phi_0}{\partial r \partial z} - \nu \frac{\partial^2 \eta}{\partial z^2} \frac{\partial \phi_0}{\partial r}. \quad (4.14)$$

Noting that

$$\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} \equiv \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial}{\partial r} \right), \quad (4.15)$$

we can integrate to find the solution at each order

$$\phi_n = \int \frac{1}{r} \left( \int r \phi_{n-1} dr \right) dr + A_n(z, t) \ln r + B_n(z, t), \quad \text{for } n = 0, 1, 2, \quad (4.16)$$

where  $\phi_{-1} = 0$  and  $A_n(z, t)$ ,  $B_n(z, t)$  are unknown functions of integration which generally depend on  $z$  and  $t$ . The function  $B_n(z, t)$  is an unknown that will govern the weakly nonlinear evolution equation for a Plateau border. Let us denote  $B_0(z, t) \equiv B(z, t)$ . With  $\phi_{-1} = 0$  and the recurrence relations (4.16), after some integration and application of the boundary conditions (4.9)–(4.11), we find

$$\phi_0(z, t) = B(z, t), \quad (4.17)$$

$$\phi_1(r, z, t) = \left( -\frac{r^2}{4} + \frac{v^2}{2} \ln r \right) \frac{\partial^2 B}{\partial z^2} + D(z, t), \quad (4.18)$$

$$\begin{aligned} \phi_2(r, z, t) = & \left( \frac{r^4}{64} - \frac{v^2}{8} [r^2(\ln r - 1)] + \ln r \left( \frac{v^4 \ln v}{4} - \frac{3v^4}{16} \right) \right) \frac{\partial^4 B}{\partial z^4} \\ & + v^2 \eta(z, t) \frac{\partial^2 B}{\partial z^2} \ln r + \frac{\partial^2 D}{\partial z^2} \left( -\frac{r^2}{4} + \frac{v^2}{2} \ln r \right) + F(z, t), \end{aligned} \quad (4.19)$$

where  $B(z, t)$ ,  $D(z, t)$ ,  $F(z, t)$  are arbitrary functions of integration. Following Wang *et al.* (2015), we introduce the canonical KdV scaling

$$\xi = z - Ct, \quad \tau = \epsilon^2 t, \quad (4.20)$$

where  $C$  is the wave speed that is to be determined. The derivatives with respect to  $z$  and  $t$  in the equations of motion are then rewritten in terms of the derivative with respect to  $\xi$  and  $\tau$  using the transformations

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial \xi}, \quad \frac{\partial}{\partial t} = -C \frac{\partial}{\partial \xi} + \epsilon^2 \frac{\partial}{\partial \tau}. \quad (4.21)$$

Substituting the change of variables (4.20) with the transformations (4.21) and the expansion for  $R(z, t)$ , (4.7), into the Bernoulli (4.4), we find the Bernoulli equation becomes

$$\begin{aligned} & -C\epsilon^2 \frac{\partial \phi}{\partial \xi} - \epsilon^4 \frac{\partial \phi}{\partial \tau} + \frac{1}{2} \left[ \left( \frac{\partial \phi}{\partial r} \right)^2 + \epsilon^6 v^2 \left( \frac{\partial \eta}{\partial \xi} \frac{\partial \phi}{\partial r} \right)^2 + \epsilon^2 \left( \frac{\partial \phi}{\partial \xi} \right)^2 + 2\epsilon^4 v \frac{\partial \eta}{\partial \xi} \frac{\partial \phi}{\partial r} \frac{\partial \phi}{\partial \xi} \right] \\ & + \frac{(1-v)\epsilon^6 \frac{\partial^2 \eta}{\partial \xi^2} (1 + \epsilon^2 \eta(\xi, \tau)) - \epsilon^2 - (v-1)^2 \epsilon^8 \left( \frac{\partial \eta}{\partial \xi} \right)^2}{(1 + \epsilon^2 \eta(\xi, \tau)) \sqrt{1 + (v-1)^2 \epsilon^6 \left( \frac{\partial \eta}{\partial \xi} \right)^2}} + \epsilon^2 = 0 \end{aligned}$$

on  $r = 1 + \epsilon^2 \eta(\xi, \tau)$ . (4.22)

Similarly, the kinematic condition (4.5) becomes

$$-\epsilon^4 \mathcal{C} \frac{\partial \eta}{\partial \xi} (1 - \nu) + \epsilon^6 \frac{\partial \eta}{\partial \tau} (1 - \nu) - \frac{\partial \phi}{\partial r} + \epsilon^2 \frac{\partial \eta}{\partial \xi} (1 - \nu) \left( \nu \epsilon^4 \frac{\partial \eta}{\partial \xi} \frac{\partial \phi}{\partial r} + \epsilon^2 \frac{\partial \phi}{\partial \xi} \right) = 0 \quad \text{on } r = 1 + \epsilon^2 \eta(\xi, \tau). \quad (4.23)$$

We also note the asymptotic solution to Laplace's equation in the new variables  $(\xi, \tau)$  is given by

$$\phi_0(\xi, \tau) = B(\xi, \tau), \quad (4.24)$$

$$\phi_1(r, \xi, \tau) = \left( -\frac{r^2}{4} + \frac{\nu^2}{2} \ln r \right) \frac{\partial^2 B}{\partial \xi^2} + \theta_2(\xi, \tau), \quad (4.25)$$

$$\begin{aligned} \phi_2(r, \xi, \tau) = & \left( \frac{r^4}{64} - \frac{\nu^2}{8} [r^2 (\ln r - 1)] + \ln r \left( \frac{\nu^4 \ln \nu}{4} - \frac{3\nu^4}{16} \right) \right) \frac{\partial^4 B}{\partial \xi^4} \\ & + \nu^2 \eta(z, t) \frac{\partial^2 B}{\partial \xi^2} \ln r + \frac{\partial^2 D}{\partial \xi^2} \left( -\frac{r^2}{4} + \frac{\nu^2}{2} \ln r \right) + \theta_4(\xi, \tau), \end{aligned} \quad (4.26)$$

where we denote

$$\theta_2(\xi, \tau) \equiv D(z, t), \quad \theta_4(\xi, \tau) \equiv F(z, t). \quad (4.27)$$

We expand in Taylor series about  $r = 1$  the Bernoulli condition (4.22), to move the terms in the equation evaluated at the unknown free surface  $r = 1 + \epsilon^2 \eta(\xi, \tau)$  to the known location of  $r = 1$ . Then substituting the perturbation expansion for the velocity potential (4.8) yields, at orders  $O(\epsilon^4)$  and  $O(\epsilon^6)$ , respectively,

$$-\mathcal{C} \frac{\partial \phi_0}{\partial \xi} \Big|_{r=1} + \eta = 0, \quad (4.28)$$

and

$$-\mathcal{C} \frac{\partial \phi_1}{\partial \xi} \Big|_{r=1} + \frac{\partial \phi_0}{\partial \tau} \Big|_{r=1} + \frac{1}{2} \left( \frac{\partial \phi_0}{\partial \xi} \Big|_{r=1} \right)^2 - \eta^2 + (1 - \nu) \frac{\partial^2 \eta}{\partial \xi^2} = 0. \quad (4.29)$$

Similarly, for the kinematic condition we obtain at orders  $O(\epsilon^4)$  and  $O(\epsilon^6)$ , respectively,

$$-\mathcal{C} \frac{\partial \eta}{\partial \xi} (1 - \nu) - \frac{\partial \phi_1}{\partial r} \Big|_{r=1} = 0, \quad (4.30)$$

and

$$\frac{\partial \eta}{\partial \tau} (1 - \nu) - \eta \frac{\partial^2 \phi_1}{\partial r^2} \Big|_{r=1} - \frac{\partial \phi_2}{\partial r} \Big|_{r=1} + (1 - \nu) \frac{\partial \eta}{\partial \xi} \frac{\partial \phi_0}{\partial \xi} \Big|_{r=1} = 0. \quad (4.31)$$

#### 4.3. Leading-order solution $O(\epsilon^4)$

Upon substituting the asymptotic solutions to the Laplace equations (4.24)–(4.26) into the Bernoulli condition (4.28) at  $O(\epsilon^4)$ , we obtain

$$\mathcal{C} \frac{\partial \eta}{\partial \xi} = \frac{\nu + 1}{2} \frac{\partial^2 B}{\partial \xi^2}. \quad (4.32)$$



Similarly, for the kinematic condition at  $O(\epsilon^4)$  (4.30), we obtain

$$-C \frac{\partial B}{\partial \xi} + \eta = 0. \quad (4.33)$$

Eliminating  $B(\xi, \tau)$  between (4.32) and (4.33), we find

$$C^2 = \frac{\nu + 1}{2}. \quad (4.34)$$

This expression for  $C$  represents the long-wave speed. As expected, the value of  $C$  agrees with long-wave speed (3.22) from linear theory in § 3.

#### 4.4. First-order solution at $O(\epsilon^6)$

Proceeding similarly to the derivation of the leading-order solution, we substitute the solutions to the Laplace equations (4.24)–(4.26) into the Bernoulli (4.29) and kinematic conditions (4.31) at  $O(\epsilon^6)$ , to obtain

$$\frac{\partial B}{\partial \tau} - \eta^2 + (1 - \nu) \frac{\partial^2 \eta}{\partial \xi^2} + \frac{1}{2} \left( \frac{\partial B}{\partial \xi} \right)^2 + \frac{C}{4} \frac{\partial^3 B}{\partial \xi^3} - C \frac{\partial \theta_2}{\partial \xi} = 0, \quad (4.35)$$

$$\begin{aligned} \frac{\partial \eta}{\partial \tau} (1 - \nu) + \eta \left( \frac{1 - \nu^2}{2} \frac{\partial^2 B}{\partial \xi^2} \right) - \frac{\partial^4 B}{\partial \xi^4} \left( \frac{1}{16} + \frac{\nu^2}{8} + \frac{\nu^4 \ln \nu}{4} - \frac{3\nu^4}{16} \right) \\ + \frac{1 - \nu^2}{2} \frac{\partial^2 \theta_2}{\partial \xi^2} + (1 - \nu) \frac{\partial \eta}{\partial \xi} \frac{\partial B}{\partial \xi} = 0, \end{aligned} \quad (4.36)$$

respectively. Differentiating (4.35) with respect to  $\xi$ , we find

$$\frac{\partial^2 B}{\partial \xi \partial \tau} - 2\eta \frac{\partial \eta}{\partial \xi} + (1 - \nu) \frac{\partial^3 \eta}{\partial \xi^3} + \frac{\partial^2 B}{\partial \xi^2} \frac{\partial B}{\partial \xi} + \frac{C}{4} \frac{\partial^4 B}{\partial \xi^4} - C \frac{\partial^2 \theta_2}{\partial \xi^2} = 0. \quad (4.37)$$

We can now eliminate the unknown arbitrary function of integration  $\theta_2(\xi, \tau)$  from (4.36) and (4.37). Then, substituting the leading-order solution, namely:  $\partial B / \partial \xi = \eta / C$  into the resulting equation, we obtain

$$\frac{\partial \eta}{\partial \tau} + \alpha \eta \frac{\partial \eta}{\partial \xi} + \beta \frac{\partial^3 \eta}{\partial \xi^3} = 0, \quad (4.38)$$

where

$$\alpha = \frac{3 - \nu}{4C} \approx 0.44446, \quad (4.39)$$

$$\beta = \frac{1}{8C(\nu - 1)} \left( \nu^4 \ln \nu - \frac{3}{4} \nu^4 - 2\nu^3 + 3\nu^2 + 2\nu - \frac{9}{4} \right) \approx -0.075835, \quad (4.40)$$

$$C^2 = \frac{\nu + 1}{2} \approx 1.0774. \quad (4.41)$$

The coefficients of the nonlinear and dispersive terms in the evolution equation for a Plateau border (4.38) are given to five significant figures. Equation (4.38) is the single evolution equation for the waves travelling on the Plateau border and is a form of the KdV equation.

The solutions of (4.38) are well-known and can be calculated analytically using the substitution  $\eta = (3s/\alpha) \text{sech}^2 \vartheta$ , to yield

$$\eta(\xi, \tau) = \frac{3s}{\alpha} \text{sech}^2 \left( \frac{1}{2} \sqrt{\frac{s}{\beta}} (\xi - s\tau) \right), \quad (4.42)$$

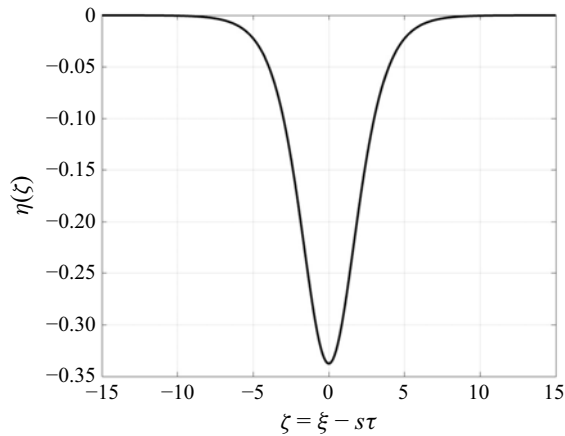


Figure 5. Plot of solution (4.42) to the KdV (4.38) for  $s = -0.05$ .

where  $s = \alpha\eta_0/3$  and  $\eta_0$  is the amplitude of the wave. Based on the coefficients of the nonlinear and dispersive terms,  $\alpha$  and  $\beta$ , the evolution (4.38) only admits depression solitary waves with  $s < 0$ . This means a solitary wave travels with speed slightly less than the long-wave speed, i.e.  $(C + s\epsilon^2)$ . A plot of the solution (4.42) for  $s = -0.05$  is depicted in figure 5.

## 5. Discussion and conclusions

We have analysed the linear stability of a uniform flow and examined the weakly nonlinear evolution of long-wavelength disturbances along a Plateau border, by generalising the azimuthal averaging technique devised by Argentina *et al.* (2015). Our linear stability analysis found that the basic flow is stable for all small perturbations, and waves emerge on the free surface. It is interesting to note that our linear theory predicts no instabilities. We established that there is only one point along the dispersion curve where solitary waves can bifurcate, the long-wave limit. An asymptotic analysis analogous to that used to derive the KdV equation for water waves (Vanden-Broeck 2010) was employed, leading to the KdV equation (4.38) for waves travelling along a Plateau border. Finally, we observe that our KdV equation admits only depression solitary waves for the values of nonlinear and dispersive terms present. These waves travel with a speed slightly less than the long-wave limit, which is in agreement with the observations of Bouret *et al.* (2016).

It would be interesting to validate this KdV equation (4.38) with the full nonlinear problem, as well as consider the effects of  $\theta$ -dependence on the linear stability of a Plateau border and investigate the linear stability of a Plateau border from the Navier–Stokes equations. These aspects are left for future work.

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## Appendix A

The differential transformations that map the first-order partial derivatives in Cartesian coordinates to  $(r, \theta, z)$  coordinate system for a Plateau border are

$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z^*} \end{pmatrix} = \mathbf{J} \begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} \end{pmatrix}, \quad (\text{A1})$$

and Jacobian matrix,  $\mathbf{J}$ , of this transformation from Cartesian coordinates to a coordinate that describes a Plateau border is

$$\mathbf{J} = \begin{pmatrix} \cos \theta & -\frac{\sin \theta}{r} & 0 \\ -\sin \theta & -\frac{\cos \theta}{r} & 0 \\ \nu \sin \theta \frac{\partial R}{\partial z} & \frac{\nu \cos \theta}{r} \frac{\partial R}{\partial z} & 1 \end{pmatrix}, \quad (\text{A2})$$

where we note that  $\det(\mathbf{J}) = -1/r \neq 0$  anywhere in the domain of the Plateau border. Therefore, the coordinate transformation is proper one, in the sense that is invertible everywhere in the domain of the Plateau border. The corresponding inverse relations to (A1) are

$$\begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z^*} \end{pmatrix}, \quad (\text{A3})$$

where

$$\mathbf{J}^{-1} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ -r \sin \theta & -r \cos \theta & 0 \\ 0 & \nu \frac{\partial R}{\partial z} & 1 \end{pmatrix}, \quad (\text{A4})$$

defines the inverse Jacobian such that  $\mathbf{J}\mathbf{J}^{-1} = \mathbf{I}$ , where  $\mathbf{I}$  is the identity matrix.

Note the divergence of a general vector  $\mathbf{v}(\theta, z) = v_1(\theta, z)\hat{\mathbf{i}} + v_2(\theta, z)\hat{\mathbf{j}} + v_3(\theta, z)\hat{\mathbf{k}}$ , from (A1), is given by

$$\begin{aligned} \nabla \cdot \hat{\mathbf{v}} &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \\ &= \left( \cos \theta \frac{\partial v_1}{\partial r} - \frac{\sin \theta}{r} \frac{\partial v_1}{\partial \theta} \right) + \left( -\sin \theta \frac{\partial v_1}{\partial r} - \frac{\cos \theta}{r} \frac{\partial v_2}{\partial \theta} \right) \\ &\quad + \left( \nu \sin \theta \frac{\partial R}{\partial z} \frac{\partial v_3}{\partial r} + \frac{\nu \cos \theta}{r} \frac{\partial v_3}{\partial \theta} + \frac{\partial v_3}{\partial z} \right). \end{aligned} \quad (\text{A5})$$

Since  $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z\}$  from (2.14)–(2.16) forms a non-orthogonal coordinate system, we can use the dual basis,  $\mathbf{e}^j$  to determine the components of the fluid velocity. These basis vectors are defined by  $\mathbf{e}_i \cdot \mathbf{e}^j = \delta_i^j$  (the Kronecker delta). For any coordinate system

(whether orthogonal or not), there are two natural ways to define a set of basis vectors: tangential to coordinate curves (covariant basis) or perpendicular to the coordinate curves (contravariant basis). The distinction between the covariant and contravariant basis vectors does not arise for orthogonal coordinate systems, because in such systems  $\mathbf{e}^j$  is simply  $\mathbf{e}_j$ , which by definition satisfies:  $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$ . The dual basis can be found by solving the following linear system of equations:

$$(\mathbf{e}_i)_j \cdot (\mathbf{e}^k)_j = \delta_{ik}, \quad (\text{A6})$$

where  $(\mathbf{e}_i)_j$  denotes that  $j$ th component of  $\mathbf{e}_i$  basis vector. Hence, we find the dual basis vectors to be

$$\mathbf{e}^1 \equiv \mathbf{e}^r = \cos \theta \hat{\mathbf{i}} - \sin \theta \hat{\mathbf{j}} + \frac{\partial R}{\partial z} \nu \sin \theta \hat{\mathbf{k}}, \quad (\text{A7})$$

$$\mathbf{e}^2 \equiv \mathbf{e}^\theta = -\frac{\sin \theta}{r} \hat{\mathbf{i}} - \frac{\cos \theta}{r} \hat{\mathbf{j}} + \frac{\partial R}{\partial z} \frac{\nu \cos \theta}{r} \hat{\mathbf{k}}, \quad (\text{A8})$$

$$\mathbf{e}^3 \equiv \mathbf{e}^z = \hat{\mathbf{k}}. \quad (\text{A9})$$

Taking the scalar product of the fluid velocity vector  $\mathbf{u}$  with  $\mathbf{e}^r$ , we find the  $u^r$  component of the velocity field, since  $\mathbf{u} \cdot \mathbf{e}^r = u^r (\mathbf{e}_r \cdot \mathbf{e}^r) = u^r$ . The following results:

$$\mathbf{e}_r \cdot \hat{\mathbf{i}} = \cos \theta, \quad \mathbf{e}_r \cdot \hat{\mathbf{j}} = -\sin \theta, \quad (\text{A10})$$

$$\mathbf{e}_\theta \cdot \hat{\mathbf{i}} = -r \sin \theta, \quad \mathbf{e}_\theta \cdot \hat{\mathbf{j}} = -r \cos \theta, \quad (\text{A11})$$

$$\mathbf{e}_z \cdot \hat{\mathbf{i}} = 0, \quad \mathbf{e}_z \cdot \hat{\mathbf{j}} = \nu \frac{\partial R}{\partial z}, \quad (\text{A12})$$

can be used to obtain the symmetric conditions (2.32).

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